

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Discuss in detail any three techniques for collection of gaseous pollutant.
- Q.37 Explain in Detail Dry scrubbing and wet scrubbing method for desulphurization of flue gases.
- Q.38 Discuss the Secondary treatment of wastewater in detail.

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Chemical Engineering, Chem (P&P), Found. & Forg.
Subject : Pollution Control in
Chemical Process Industry/ Env. Engg. & Safety

SECTION-A

Q.1 Air Pollution can be classified as

- Natural
- Primary
- Gaseous & Particulate
- All of the above

Q.3 Which of the following is the source of water pollution.

a) Industrial Waste b) Domestic Sewage

c) Municipal Waste d) All of the above

Q.4 Which of the following technique is used to remove sulphur from fuel.

- a) Electrostatic Precipitation
- b) Thermal Precipitation
- c) Absorption in liquid
- d) Desulphurization

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- Q.5 Which of the following is a gaseous air pollutant.
- Smoke
 - Fly ash
 - CO₂
 - Dust
- Q.6 Which instrument is used for particulate collection.
- ESP
 - pH meter
 - Colorimeter
 - All of these
- Q.7 Which of the following is secondary treatment method of waste water.
- Sedimentation
 - Activated sludge process
 - Floatation
 - None of these
- Q.8 NDIR Technique is used for measurement of
- Nitrogen oxide
 - Sulphur oxide
 - Carbon monoxide
 - Hydrocarbon
- Q.9 Which of the following is the technique for solid waste disposal.
- Sedimentation
 - Impingement
 - Thermal precipitation
 - Composting
- Q.10 Which of the following is a method to control gaseous pollutants.
- Adsorption by solid
 - Sedimentation
 - ESP
 - Filtration

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1 = 10)

- Q.11 Name two types of sources of Air Pollution.
- Q.12 Name two air pollutants.
- Q.13 In _____, the sample is collected by filling an evacuated flask or on inflatable bag.
- Q.14 Name secondary water treatment.
- Q.15 Name two solid waste.
- Q.16 Define COD.
- Q.17 Name two water pollutants.
- Q.18 Write any two types of water pollutant.
- Q.19 Define the term pollution.
- Q.20 Name man made sources of pollutants.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5 = 60)

- Q.21 Describe the different scales for measurement of Air pollution.
- Q.22 Describe the chemiluminescent analyzer for the measurement of Nitrogen oxide (with diagram).
- Q.23 Explain the technique used for the measurement of carbon oxides in atmosphere.
- Q.24 Discuss about the equipment modification of existing equipment for control of Air Pollution.
- Q.25 Describe the venturi scrubber and Fabric filter technique for particulate control.

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